

Financial Time Series

46-929

Homework #6

Due: Friday, May 7, 2021, 1:30pm.

1. Download from the canvas data folder fredgraph.csv. This file contains data from the St. Louis Federal Reserve giving the U.S. Treasury monthly average constant maturity rates for 1 year, 5 years and 10 years dating back to 1953. Plot these variables, perform a Dickey Fuller test to check for stationarity. If needed, difference the data to transform it to $\mathcal{I}(1)$. Perform cointegration tests (Johansen, Engle-Granger and Phillips-Ouliaris) to check if any pairs of these data are cointegrated, following what was done in class or equivalently follow the example give in
2. Consider a univariate AR(p) process

$$Y_t - \mu = \phi_1(Y_{t-1} - \mu) + \cdots \phi_p(Y_{t-p} - \mu) + \epsilon_t.$$

Write this in state space form, identifying all of the components of that form taking $\mathbf{X}_t = (Y_t - \mu, \dots, Y_{t+1-p} - \mu)$ (p-dimensional) and Y_t (1-dimensional) as the variable in the observation equation.

3. Consider the local trend model

$$M_{t+1} = M_t + V_t,$$

$$Y_t = M_t + W_t,$$

where $\{V_t\}$ is $\text{WN}(0, \sigma_V^2)$ and $\{W_t\}$ is $\text{WN}(0, \sigma_W^2)$

- a) Show that Y_t can be written as an ARIMA(0,1,1) model

$$Y_t = Y_{t-1} + Z_t + \theta Z_{t-1},$$

for an appropriately chosen noise sequence Z_t , where $\{Z_t\}$ is $\text{WN}(0, \sigma^2)$.

- b) Determine possible solutions for θ and σ in terms of σ_V^2 and σ_W^2 .
- c) Let $\sigma_W^2 = 8$ and $\sigma_V^2 = 20$, simulate a series of Y_t according to this local trend model for the possible solutions found in part b) with these specific values for σ_W^2 and σ_V^2 .
- d) Perform a time series analysis on the simulated Y_t series. What do you conclude?